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Q1. Smart University Payroll Management System

Score: 25.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

class Main {
    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);

        int n = Integer.parseInt(s.nextLine());
        s.nextLine();

        Employee[] emp = new Employee[n];

        for (int i = 0; i < n; i++) {
            String type = s.nextLine();
            int id = Integer.parseInt(s.nextLine());
            String name = s.nextLine();
            String dept = s.nextLine();
            double basicsalary = Double.parseDouble(s.nextLine());

            if (type.equalsIgnoreCase("TeachingFaculty")) {
                int teachingHour = Integer.parseInt(s.nextLine());
                double incentiveperHour = Double.parseDouble(s.nextLine());
                s.nextLine();
                emp[i] = new TeachingFaculty(id, name, dept, basicsalary, teachingHour, incentiveperHour);
            } else if (type.equalsIgnoreCase("TechnicalStaff")) {
                int overtimeHour = Integer.parseInt(s.nextLine());
                double overtimeRate = Double.parseDouble(s.nextLine());
                s.nextLine();
                emp[i] = new TechnicalStaff(id, name, dept, basicsalary, overtimeHour, overtimeRate);
            } else if (type.equalsIgnoreCase("VisitingFaculty")) {
                int lecturehandled = Integer.parseInt(s.nextLine());
                double paymentperHour = Double.parseDouble(s.nextLine());
                s.nextLine();
                emp[i] = new VisitingFaculty(id, name, dept, basicsalary, lecturehandled, paymentperHour);
            }
        }

        String s1 = s.nextLine();
        int searchID = Integer.parseInt(s.nextLine());

        System.out.println("Employee Details");
        System.out.println();

        double total = 0;
        double maxsalary = -1;
        Employee highest = null;

        for (Employee e : emp) {
            e.display();
            double salary = e.calculatesalary();
            System.out.println("Monthly Salary : " + salary);
            System.out.println();
            total += salary;

            if (salary > maxsalary) {
                maxsalary = salary;
                highest = e;
            }
        }

        System.out.println("Highest Paid Employee");
```

```
        System.out.println("Employee ID : " + highest.getid());
        System.out.println("Salary : " + highest.calculatesalary());
        System.out.println();

        System.out.println("Average Salary");
        System.out.printf("%.2f\n", total / n);
        System.out.println();

        boolean found = false;
        for (Employee e : emp) {
            if (searchID == e.getid()) {
                System.out.println("Employee Found");
                System.out.println("Employee ID : " + e.getid());
                System.out.println("Name : " + e.getname());
                System.out.println("Salary : " + e.calculatesalary());
                found = true;
                break;
            }
        }

        if (!found) {
            System.out.println("Employee Not Found");
        }
    }
}

class Employee {
    private String emp_name;
    private String department;
    private int emp_id;
    private double basicsalary;

    Employee(int emp_id, String emp_name, String department, double basicsalary) {
        this.emp_id = emp_id;
        this.emp_name = emp_name;
        this.department = department;
        this.basicsalary = basicsalary;
    }

    public String getname() {
        return emp_name;
    }

    public String getdept() {
        return department;
    }

    public int getid() {
        return emp_id;
    }

    public double getbasicsalary() {
        return basicsalary;
    }

    public double calculatesalary() {
        return basicsalary;
    }

    void display() {
        System.out.println("Employee ID : " + emp_id);
        System.out.println("Name : " + emp_name);
        System.out.println("Department : " + department);
    }
}
```

```
}

class TeachingFaculty extends Employee {
    int teachingHour;
    double incentiveperHour;

    TeachingFaculty(int id, String name, String dept, double basicsalary, int teachingHour, double incentiveperHour) {
        super(id, name, dept, basicsalary);
        this.teachingHour = teachingHour;
        this.incentiveperHour = incentiveperHour;
    }

    public double calculatessalary() {
        return getbasicsalary() + (teachingHour * incentiveperHour);
    }
}

class TechnicalStaff extends Employee {
    int overtimeHour;
    double overtimeRate;

    TechnicalStaff(int id, String name, String dept, double basicsalary, int overtimeHour, double overtimeRate) {
        super(id, name, dept, basicsalary);
        this.overtimeHour = overtimeHour;
        this.overtimeRate = overtimeRate;
    }

    public double calculatessalary() {
        return getbasicsalary() + (overtimeHour * overtimeRate);
    }
}

class VisitingFaculty extends Employee {
    int lecturehandled;
    double paymentperHour;

    VisitingFaculty(int id, String name, String dept, double basicsalary, int lecturehandled, double paymentperHour) {
        super(id, name, dept, basicsalary);
        this.lecturehandled = lecturehandled;
        this.paymentperHour = paymentperHour;
    }

    public double calculatessalary() {
        return lecturehandled * paymentperHour;
    }
}
```

INPUT

```
3

TeachingFaculty
101
Dr.Arun
CSE
50000
20
500

TechnicalStaff
102
Mr.Ravi
ECE
35000
15
300

VisitingFaculty
103
Mr.Kumar
IT
0
```

40
1200

Search Employee ID:
102

OUTPUT

Employee Details

Employee ID : 101
Name : Dr.Arun
Department : CSE
Monthly Salary : 60000.0

Employee ID : 102
Name : Mr.Ravi
Department : ECE
Monthly Salary : 39500.0

Employee ID : 103
Name : Mr.Kumar
Department : IT
Monthly Salary : 48000.0

Highest Paid Employee
Employee ID : 101
Salary : 60000.0

Average Salary
49166.67

Employee Found
Employee ID : 102
Name : Mr.Ravi
Salary : 39500.0

Q2. Smart Vehicle Insurance Claim System

Score: 0.00 TC: 0/3 passed

CODE

```
import java.util.*;

abstract class Vehicle {
    private String vehicleNumber;
    private String ownerName;
    private double insuredAmount;

    public Vehicle(String vehicleNumber, String ownerName, double insuredAmount) {
        this.vehicleNumber = vehicleNumber;
        this.ownerName = ownerName;
        this.insuredAmount = insuredAmount;
    }

    public String getVehicleNumber() {
        return vehicleNumber;
    }

    public String getOwnerName() {
        return ownerName;
    }

    public double getInsuredAmount() {
        return insuredAmount;
    }

    public abstract double calculateClaimAmount();
}

class Car extends Vehicle {
    public Car(String vehicleNumber, String ownerName, double insuredAmount) {
        super(vehicleNumber, ownerName, insuredAmount);
    }

    @Override
    public double calculateClaimAmount() {
        return getInsuredAmount() * 0.80;
    }
}

class Bike extends Vehicle {
    public Bike(String vehicleNumber, String ownerName, double insuredAmount) {
        super(vehicleNumber, ownerName, insuredAmount);
    }

    @Override
    public double calculateClaimAmount() {
        return getInsuredAmount() * 0.60;
    }
}

class Truck extends Vehicle {
    public Truck(String vehicleNumber, String ownerName, double insuredAmount) {
        super(vehicleNumber, ownerName, insuredAmount);
    }

    @Override
    public double calculateClaimAmount() {
        return getInsuredAmount() * 0.90;
    }
}
```

```
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        sc.nextLine(); // consume newline

        Vehicle[] vehicles = new Vehicle[n];

        for (int i = 0; i < n; i++) {
            String type = sc.nextLine().trim();
            String vehicleNumber = sc.nextLine().trim();
            String ownerName = sc.nextLine().trim();
            double insuredAmount = Double.parseDouble(sc.nextLine().trim());

            if (type.equalsIgnoreCase("Car")) {
                vehicles[i] = new Car(vehicleNumber, ownerName, insuredAmount);
            } else if (type.equalsIgnoreCase("Bike")) {
                vehicles[i] = new Bike(vehicleNumber, ownerName, insuredAmount);
            } else {
                vehicles[i] = new Truck(vehicleNumber, ownerName, insuredAmount);
            }
        }

        String searchVehicleNumber = sc.nextLine().trim();

        double highestClaim = -1;
        Vehicle highestVehicle = null;

        for (Vehicle v : vehicles) {
            double claim = v.calculateClaimAmount();
            System.out.println("Vehicle : " + v.getVehicleNumber());
            System.out.println("Claim Amount : " + claim);

            if (claim > highestClaim) {
                highestClaim = claim;
                highestVehicle = v;
            }
        }

        System.out.println("Highest Claim Vehicle");
        System.out.println(highestVehicle.getVehicleNumber());
        System.out.println(highestClaim);

        boolean found = false;

        for (Vehicle v : vehicles) {
            if (v.getVehicleNumber().equalsIgnoreCase(searchVehicleNumber)) {
                System.out.println("Vehicle Found");
                System.out.println("Owner : " + v.getOwnerName());
                System.out.println("Claim Amount : " + v.calculateClaimAmount());
                found = true;
                break;
            }
        }

        if (!found) {
            System.out.println("Vehicle Not Found");
        }

        sc.close();
    }
}
```

INPUT

```
3
Car
TN10AB1234
Ramesh
800000
Bike
TN09XY5678
Suresh
100000
Truck
TN11TR9999
Logan
1500000
Search Vehicle:
TN09XY5678
```

OUTPUT

```
Compilation Error:
C:\Windows\TEMP\sandbox_192472152RD_6a68d2917de456.64782434\Vehicle.java:62: error: class Main is public, should be declared
in a file named Main.java
public class Main {
      ^
1 error
```